falsify — the board

Every phase complete. Ten gates green, ten invariants held, and a strategy the engine declines to trade -- which is the machinery working, not failing.

Commit 2efa5ef · 2026-08-28 · 475/495 tests collected (20 deselected) in 6.53s

The finding

PLAYBOOK asks for one thing above the rest: *“If your alpha t-stat drops below 2 after controlling for momentum, say so in the README. That single act of intellectual honesty is worth more to a reader than a 2.5 Sharpe.”*

It does not drop below 2. It drops to **zero** — t = -0.01. The +0.606 annualised Sharpe `TimeSeriesMomentum(12m,1m)` earns on SPY over 2015-2024 is fully accounted for by two exposures the regression names: **+0.618 on the market**, because a trend follower is long a rising index most of the decade, and **+0.244 on UMD**, because it is a momentum strategy and UMD is the momentum factor. Price both and nothing remains.

Carhart four-factor attribution, HAC standard errors, close-to-close

strategy	alpha/yr	HAC t	R2	Mkt-RF	SMB	HML	UMD
BuyAndHold	+0.12%	+0.37	0.995	+0.975	-0.126	+0.015	-0.007
TSMomentum(12m,1m)	-0.02%	-0.01	0.408	+0.618	-0.064	+0.182	+0.244
TSMomentum(12m,3m)	-6.43%	-1.47	0.370	+0.575	-0.089	+0.094	+0.284
XS momentum 12m	-2.51%	-1.29	0.520	+0.006	+0.032	-0.020	+0.323
XS momentum 6m	-0.60%	-0.30	0.366	-0.026	+0.010	+0.017	+0.259
XS momentum 1m	-5.50%	-2.49	0.056	-0.048	-0.015	-0.023	+0.065

The buy-and-hold row is the calibration that makes the rest readable: SPY loads 0.975 on the market at R^2 0.995 with no alpha, which is exactly what an index fund should show. Run against the `next_open` convention that same check gave β = 0.367 and R^2 = 0.159 — because `next_open` measures open-to-open while the factors are close-to-close, and the two correlate 0.40 daily. Every β in the first table was wrong and nothing else would have flagged it.

Every layer agrees, and they were built independently

The strongest check available is not any one number but the fact that seven independent measurements of the same strategy reach the same verdict. Run on real SPY through the whole chain:

layer	measured	verdict
B3 trials ledger	24 trials for 24 configurations, idempotent on rerun	counted
B4 effective N	N = 24 -> N_eff 2.21, compression 0.09	the search was narrow
B5 bootstrap CI	+0.575 [-0.039, +1.276] 95%	contains zero
B1 HAC t-statistic	+1.922	below 2
G9 PBO	0.8214 over 24 configurations	above the 0.5 line
Phase 7 alpha	-0.02%/yr at t = -0.01	no alpha
01 Part D contract	interpretable = False, ships = False	does not ship

None of these was tuned to agree with the others. The bootstrap knows nothing about the factor model; PBO knows nothing about the sample length; the ledger counts configurations without looking at their returns. That they converge is the argument.

Phase 7 — the long/short spread

Nine SPDR sector funds, dollar-neutral tertile long/short, 2015-2024, zero cost:

construction	`SR`	`+/-SE`	`turns/yr`	`HAC t`
XS momentum 12m, monthly	`-0.065`	`0.334`	`4.79`	`-0.20`
XS momentum 12m, daily	`-0.023`	`0.334`	`23.03`	`-0.07`
XS momentum 6m, monthly	`+0.157`	`0.325`	`6.58`	`+0.51`
XS momentum 1m, monthly	`-0.432`	`0.318`	`16.31`	`-1.37`

No edge, and that is the result — nothing approaches a HAC t of 2. The gap from the time-series case is the content: the same signal earns +0.606 on SPY and nothing at all when held dollar-neutral. That difference is the market's drift, which a long-biased strategy collects and a neutral book cannot.

The panel engine is Part E written a second time, so it is held to G2's standard: at N = 1 it reproduces `run_vectorized` bitwise, max |diff| = 0.000e+00. That check caught a real bug on its first run — the warm-up was `max(first_nonzero, lag)` instead of `first_nonzero + lag`, wrong by 1,313 on a 10,000 account.

Checklists

Every gate, invariant and specification document against what exists at this commit.

Gate set — PLAYBOOK Part 2

item	state	note
G1 causality	GREEN	500 taus x 20 seeds, both cut modes, four zoo strategies
G2 twin engines	GREEN	worst 0.000e+00 across 1,890 combinations
G3 analytic recovery	GREEN	worst level gap 0.90 SE, paired gap 0.00 SE
G4 zero-cost identity	GREEN	288/288 bitwise identical
G5 cost monotonicity	GREEN	12/12 sweeps monotone, break-even reported
G6 null calibration	GREEN	bounds set from a 15-world replication study
G7 leakage trap	GREEN	G1 flags the injected leak
G8 walk-forward	GREEN	purge and embargo; zero index overlap asserted
G9 overfit probability	GREEN	12,870 splits; null calibrates to 0.5 at every S
G10 reproducibility	GREEN	three figures and metrics.json byte-identical

Invariants — 03 Part B

item	state	note
B1 no network before Gate 0	GREEN	phases 0-4 offline; one deliberate fetch script
B2 every number has an error bar	GREEN	structural in Performance and MetricsReport
B3 append-only trials ledger	GREEN	required on both engines; N read, never typed
B4 strategies emit weights	GREEN	sizing and execution belong to the engine
B5 both engines identical	GREEN	changed together; G2 re-run in the same commit
B6 no bfill	GREEN	panel aligns on intersection rather than filling
B7 frozen dataclasses	GREEN	checked by health_check
B8 per-observation internally	GREEN	annualised only at the reporting boundary
B9 seeds threaded explicitly	GREEN	no global RNG; G10 depends on it
B10 non-excess kurtosis	GREEN	fisher=False, asserted against the Lo collapse

Specification documents

item	state	note
00 Gate 0.0 A/B/C	**GREEN**	exact quadrature, diff 0.00e+00; the compensation figure
00 Gate 0.1-0.5	**GREEN**	recovery, convergence, signal, degeneracy, determinism
01 B1 Sharpe, SE and HAC t	**GREEN**	Lo (2002) non-normal plus Newey-West
01 B2 / B3 PSR and DSR	**GREEN**	now fed an N read from the ledger
01 B4 effective trials	**GREEN**	participation ratio primary, clustering secondary
01 B5 stationary bootstrap	**GREEN**	twin implementations, bitwise identical
01 B6 PBO via CSCV	**GREEN**	G9
01 Part C trials ledger	**GREEN**	the trap counts 50, not 1
01 Part D reporting contract	**GREEN**	ten fields, written to metrics.json
01 Part E selection rules	**GREEN**	four rules; the PBO-vs-temperature figure
02 Parts A-H	**GREEN**	A4 ruled and verified live; manifest data contract
03 Parts B, G, H	**GREEN**	ten invariants held; Part H revisited on its own terms
PLAYBOOK Phase 6 zoo	**GREEN**	MA, mean reversion, vol target, null, TSMOM
PLAYBOOK Phase 7 cross-sectional	**GREEN**	N-asset panel, long/short, attribution
PLAYBOOK Phase 8 reporting	**GREEN**	metrics.json, tearsheet, IS/OOS parameter surface
PLAYBOOK Phase 9 writeup	**GREEN**	docs/research-note.md; README leads with the result

What is left

item	why
Nothing blocking	Phases 0 through 9 are complete. The research note is written, the README leads with the result, and every figure the specification asked for exists.
If the project continues	A longer window is the only thing that would change the verdict: 10.7 years are needed to distinguish this Sharpe from selection luck and 10.0 exist. Evaluating more configurations widens that gap rather than closing it.

Generated 2026-08-28 from 2efa5ef. Every statistic here is asserted by a test in the tree and carries the standard error it was measured with; none was rounded toward a nicer number.